



INSTITUTE OF PHYSICS

TFY4510 FYSIKK, FORDYPNINGSPROSJEKT

Beliefs and the Propagation of Misinformation in Social Networks

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21st June 2025

Abstract

Understanding and mitigating misinformation propagation and its impact in society requires models which can accurately simulate how it spreads and behaves. Epidemiological models are common, but they often encounter the problem of sacrificing generality for accuracy, or vice versa. By reviewing literature from cross-disciplinary fields studying misinformation, this project adapts and modifies a Distributed Hypothesis Testing (DHT) model, where agents update their own private and public beliefs from observations and interactions with neighbors, allowing nuanced beliefs. Erdős-Rényi and scale-free Price networks were utilized to study what impact network topology, confirmation bias and conspirators have on a system's truthfulness. Plotting the time evolution of truthfulness on Erdős-Rényi and Price networks using a basic DHT model reproduced known results, confirming the model's validity. A network's connectivity was shown to be vital for a system to reach a state of complete truthfulness, requiring a degree $k > 5$ for Erdős-Rényi networks, and for the speed with which a system converges to this state. Conspirators were found to drastically decrease truthfulness, a finding consistent with previous research. Novel to this study, the system exhibited a rapid collapse in truthfulness when the number of conspirators exceeded 5–6% of the total population. Novelty applying confirmation bias on agents' interactions with each other impacted the speed of convergence towards complete truthfulness and was found to have a significant effect on the system's rapid collapse in truthfulness. In fact, confirmation bias made the system more robust, protecting some systems' truthfulness while decreasing other systems' truthfulness. This project expands the use of the DHT model to simulate misinformation propagation. It is validated as a useful tool to model misinformation in social networks and shows a promising potential to be able to simulate systems with realistic human-like agents.

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1 Introduction

In the information age, the belief in and spread of misinformation is a threat to democratic processes, collaboration in communities, and nuanced public discourse [13][32]. Understanding how misinformation propagates, its properties, and what conditions make it propagate is an important challenge. There is an vast amount of research done on models trying to accurately simulate how misinformation spreads and behaves. This research has been focused mainly on representing misinformation as a contagion, similar to a disease, and simulating its spread using epidemiological models [16][49][51]. Due to their high adaptability, epidemiological models can incorporate or exclude various classes and introduce additional parameters, often inspired by psychology, which widens their range of use. By adapting an epidemiological model to a specific scenario, it is possible to get accurate simulation results backed by empirical data for that exact scenario [50]. While adaptable, a key challenge for epidemiological models lies in the trade-off between generality and accuracy: models that remain broadly applicable often sacrifice precision in specific scenarios, and vice versa [36]. Researching this topic, there seems to be a disconnect between the different fields studying misinformation propagation. The theory and empirical evidence in social science are not necessarily mentioned or used within models made by mathematicians, physicists, and engineers. Resulting in models where agents might lack realistic behaviors. Representing misinformation as a contagion removes some of the nuance of belief systems, and trying to keep the nuance means imposing more classes and parameters.

The Distributed Hypothesis Testing (DHT) model uses social learning and belief formation [26] to simulate how misinformation propagates in social networks [37]. It gives the network’s agents private q and public p beliefs. Using these, one can measure truthfulness τ and the conflict between one agent’s beliefs, i.e. cognitive dissonance [44]. The agents are allowed to have nuanced beliefs and through making observations and interacting with their neighbors, the beliefs will be updated. The DHT model offers a flexible foundation that directly supports the integration of realistic agent behaviors, such as confirmation bias [8], allowing for a more nuanced representation of belief formation. The DHT model, for this specific purpose, has primarily been explored in Diana Riazi’s recent PhD thesis and accompanying papers [38][39][40], making further investigation crucial.

This project aims to gain an overview of models used for simulating misinformation propagation and research on misinformation in social science. Using this overview, the DHT will be reasoned for. The next goal is to further advance the research on DHT models used for simulating misinformation propagation by implementing and validating such a model. Basic DHT simulations will be analyzed on Erdős-Rényi and Barabási-Albert networks with the Price model [14][35], analyzing how different networks will affect the results. In addition, a network’s topology and properties, such as connectivity, will affect the system’s final truthfulness and the speed of convergence towards full truthfulness. The introduction of conspirators within the system is expected to negatively impact its overall truthfulness. Investigating the dual effects of imposing confirmation bias on agent interactions, particularly in systems with conspirators, will be crucial for developing more realistic agent behaviors.

In Section 2 I will give an overview and discuss relevant literature on the topic of misinformation. Because of the cross-disciplinary nature of the subject matter, this Section will include literature from social science and psychology, literature on data analysis, models and simulations, and review studies. Section 3 is where I present the mathematical background of graph theory and specific properties of Erdős-Rényi, Barabási-Albert, and Price networks. In addition, the background of the DHT model will be provided. In Section 4 I reason for the choice of the DHT model for this project. In Section 5 I give a presentation of how the results were produced. These results are shown and discussed in Section 6. In the end, Section 7 summarizes the key points of the paper and outlines important aspects to study in the future.

2 Existing Literature

2.1 Structure

In this section I will present some of the abundant amount of research on the topic of misinformation. Studies have been done within social science, physics, mathematics, and engineering fields. The studies have different methods and focuses, including data analysis, model simulation, human psychology and behavior, and reviews of current methods. The wide range of fields involved in studying misinformation makes it a challenging research topic, as making an overview comprises all cross-disciplinary fields. Existing literature uses several names for misinformation spread, or might study specific aspects of it. Fake/real news, false or true rumors, disinformation and spread of misinformation are used synonymously. The purpose of this section is to give a clear overview of some of the existing literature and to reason for the choices of the model in this project.

In the literature section, a review of all the cross-disciplinary fields will be presented. A general overview was gathered of existing models, by both reading review studies, studies modeling misinformation propagation, and studies using data collection from social media websites. Studies were found from the early to mid 2000s, but the world (at least the western world) looks quite different now 20 years later. With the confirmation of Russian election interference in the last 10 years [32], and the rise of new social media apps together with more users of social media, the situation looks different now compared with 20 years ago. Thus, the literature was mostly limited to papers published since 2018. The assumption being that this field is subject to a lot of change over a short amount of time.

2.2 Social science and psychology and misinformation

This section contains an overview of found literature on misinformation in other fields of research, specifically social science and psychology. The spread of misinformation is an important societal problem, and its study is not confined to only social science, but is studied within physics, engineering, and mathematics. It is important to understand what misinformation is, how it spreads, how it can be stopped from

spreading, and how we humans interact with it. That is the focus of social science and psychology. Since we want to model the spread of misinformation, it is also important for us to understand these properties as well. The papers below were chosen because they shed light on different aspects of the study of misinformation.

Starbird et al. point out in their paper "Disinformation as Collaborative Work: Surfacing the Participatory Nature of Strategic Information Operations" the importance of misinformation campaigns [43]. They perform three case studies, where misinformation campaigns were used to affect public discourse. The first case is trolling operations by the Internet Research Agency in St. Petersburg on social media leading up to the 2016 American election. These operations helped polarize the debate around the hashtag #BlackLivesMatter, among other topics. The second case is a disinformation campaign made to deface the White Helmets (a volunteer humanitarian group providing search and rescue assistance and medical aid) during the Syrian Civil War. The third case is how alternative media outlets support conspiracy theory claims in crisis events, often by sowing doubt in mainstream media. They conclude that disinformation operations are participatory. Implying that even with deviant agents spreading misinformation, the organic crowd in the network also contributes to the spread. And that the operations shape the social structures which they target. So, the public discourse on social media, where people might form their opinions, contains deviant agents with aims to affect the information flow to work in their advantage. Polarization is important here, especially in the first case study, where deviant agents deliberately took advantage of the polarized political landscape of US politics. Polarization of society can undermine the public's trust in their own nation's institutions [13], and will worsen communication between two groups of different opinions.

In 2010, Nyhan and Reifler published an article documenting the backfire effect [30]. The backfire effect is when one is presented with fact-checking information debunking one's intrinsically false belief, one will discard the new piece of information and strengthen one's false belief. The backfire effect has been used to explain why misperceptions survive, even though some researchers disagree on whether it actually does, or even if it exists at all [45]. In a newer study by Nyhan, "Why the backfire effect does not explain the durability of political misperceptions," he stated that the importance of the backfire effect has been overstated [29]. Instead, he concludes that fact-checking is effective in correcting people's false opinions, but its effect fades away as time passes by, or gets overwhelmed by less accurate claims from the media. Either way, corrective information, as it is understood and used today, might not be the most effective long-term tool to combat misinformation having a lasting effect.

Instead of debunking information with fact-checking, one long-term idea is to build people's critical thinking skills such that they are themselves able to discard misinformation. This was presented by Fasce and Picó in "Science as a vaccine" [15]. They found that people with a high level of scientific literacy did not believe in conspiracy theories, and that being able to think critically is not enough, one must also have the right attitude towards it. However, they also found that teaching people how to resist falsehoods and conspiracies has a limited impact.

In "Online Fake News Opinion Spread and Belief Change" [4], Altoe et al. present

a literature review of belief change and polarization in online social networks. The authors reviewed studies of users, methodologies, and modeling, collecting studies over several domains, such as politics, health, online communities, etc. They found that individuality is the main driver behind a person's belief change. Individuality, in this instance, is the personal traits and psychological constructs that make a person oneself. One of the main factors in belief change is the emotions connected to the topic. Specifically, the emotional arousal generated plays a role in how individuals interact with information, both when accepting or denying it. The main opponent of belief change was polarization, and it was found that polarization exists over multiple domains. I.e. if two groups are polarized on topic A, they are probably similarly polarized on other topics as well. The driver behind polarization was found to be trust in others, giving people with many loyal followers power over the opinions of members in their own polarized groups. Thus, targeting influencers in polarized groups who believe in misinformation with correcting information to change their opinion might be an effective way to depolarize said group. However, Jurg et al. theorize that influencers may also be influenced from the bottom up in their own polarized group [23]. This is called audience capture, and it depicts how an influencer may be forced to be more radical just to keep the engagement of their own radical audience. Like Jurg et al. found happened with Alex Jones on Youtube in 2017-18 regarding the "Sandy Hook Hoax."

Polarization is worsened by confirmation bias. Which is the bias humans have to accept information which already aligns with our own beliefs, either by searching explicitly for confirming information or ignoring/discarding countering information [8]. Implying that a polarized group is unlikely to take in and process information which does not align with their views. Furthermore, confirmation bias is moderated, and often intensified, by the emotions of the recipient, especially on political subjects, [11]. Reinforcing the role of emotions when one is to update one's beliefs. Cognitive dissonance is the conflict a person experiences when having two ideas, thoughts, morals or actions which do not align. Such a dissonance might also be a result from outside pressure [44]. This implies that if a debate is split into two polarized groups, cognitive dissonance will incentivize non-polarized people to "join" one of the groups if the pressure from them is high enough. Polarized groups will create their own echo chambers, which are chambers where people have similar opinions and only these will be shared with the group [9].

Polarization is poison for a wide public debate of a given topic, since in a polarized debate the different groups will have difficulty communicating with each other. This might be why misinformation campaigns by outside parties try to polarize groups. Influencers, news outlets, and institutions with large and loyal followings are able to drive the public discourse. Thus, they have a large responsibility to prevent polarization from taking place. However, if the public discourse is already polarized, one must try to accurately deliver correcting information to influencers. Such that the group as a whole might be de-polarized. This might be difficult, since audience capture might force influencers to not change their opinion in order to not lose their followers.

Even then, correcting information is found to somewhat work, as the backfire effect is not prevalent in negating it. To stop the corrected information from fading away

with time, one must repeatedly correct already corrected information. To correct information in a way that works, the emotional arousal in and the individuality of the recipient must be considered as they are the main drivers behind belief change. As educating people on critical thinking and making them view it in a positive light might not have an impact. All of this is important to keep in mind and understand, as society is under attack from misinformation campaigns aiming to polarize the public discourse and indirectly undermine trust in public institutions.

2.2.1 Human behavior and individuality in Distributed Hypothesis Testing Models

Distributed Hypothesis Testing (DHT) models are often used on sensor networks [34] (which is discussed in detail in sections 2.6 and 3.2) where the agents in the system are sensors instead of humans. As humans and sensors obviously behave differently, they will have different updating rules for their respective beliefs. A sensor can be given such rules. In the literature, a Bayesian update on beliefs after receiving signals and a non-Bayesian linear consensus on the log-belief of an agent’s neighbors are often used [33]. It can then be proven mathematically that such updating rules will result in an exponentially fast convergence towards a state where all agents’ beliefs in any wrong hypotheses are zero [26].

Humans will, in general, not update their beliefs in a Bayesian way with regards to incoming signals. In fact, Roberta De Filippis et al. [10] conclude that humans will update their beliefs in a non-Bayesian way when faced with signals that contradict their own beliefs. Thus, for any system with diverse beliefs between agents, a consensus on the true hypothesis will not necessarily be reached exponentially fast.

Another aspect to consider for human agents is confirmation bias with regards to interactions with neighbors. Confirmation bias describes humans’ tendency to overlook or discount information which does not align with what they inherently believe [8]. Translating this into a network where agents interact through weighted edges, confirmation bias will cause neighboring agents with different beliefs to weight their interactions lower than with neighbors with similar beliefs.

2.3 Studies analyzing datasets and how information propagates

This section contains an overview of found literature on the data analysis of the properties of information and misinformation in online networks. By analyzing users, groups, "hashtags," trends, posts, etc. one is able to understand what kind of structures facilitate misinformation spreading through networks. This is important to create a realistic model, and if one wants to model ways to contain the spread of misinformation.

In "Fake news propagates differently from real news even at early stages of spreading," Zhao et al. study the propagation patterns of fake news and real news on Weibo in

China and X (then known as Twitter) in Japan [52]. They found that, even without knowing what the piece of news is, the propagation patterns of news will indicate whether it is fake or real. The topology of the propagation network of fake news will consist of more layers and have a larger characteristic distance between nodes in the network. This can be found after only a few hours after the news has started spreading. In contrast, a real news propagation network has a star-like topology and a lower characteristic distance between nodes.

Vicario et al. study the lifetime and cascades of news, in a study from 2016, "The spreading of misinformation online" [47]. They reviewed how Facebook users consumed scientific and conspiracy news. They found that the lifetime of both types of news is similar. However, the lifetime of science news peaks at a low cascade size and stays constant for large cascade sizes. While for conspiracy news, the lifetime increases with the cascade size. Where cascade size is the number of users who shared the piece of news. They found that the main driver of the cascade size is the homogeneity and polarization of the network. Such that a homogeneous, polarized group will create the largest cascade size, regardless of the news being scientific or conspiracy.

In "Homophily and polarization in the age of misinformation" Bessi et al. study homophily in more than one million Facebook users. Where homophily is the tendency to interact with users with similar polarization as one's own. They found strong homophily in the users studied, effectively most users were in an echo chamber. They also found that users with polarization towards conspiracy groups are more inclined to spread misinformation than users with polarization towards other groups. In fact, the more misinformation spread, the more polarized a user is towards conspiracy groups. Again, we see that in systems with misinformation, polarization is strongly correlated to its spread.

Data analysis shows that polarization facilitates misinformation spreading in online social networks. Which supports the theories presented in social science (see Section 2.2). In addition, these studies can quantitatively describe the topology of the typical networks in which misinformation thrives. These networks consist of more layers than networks corresponding to the spread of real news, and the cascade size of fake news is correlated to the lifetime of said news. Couple these findings with the fact that individuality drives belief change in a user. One can hypothesize that the more emotions a piece of fake news evokes in the recipient, the more likely that recipient is to spread it onward. Data analysis of misinformation propagation ultimately shows that misinformation is not just a problem with the content and the recipients, but also how the network is connected.

2.4 Epidemiological modeling of misinformation in literature

This section contains an overview of found literature on misinformation, and epidemiological models, including thorough analyses of a selected portion of them. These types of models are the most common models I came across when researching this topic, and they have a wide range of applications as will be explained in this section. The aim of this project is to create a model which can simulate misinformation

propagation in a social network. Before deciding on a model, it is vital to comb through the models used in existing literature. The findings are discussed in their own subsection below.

Epidemiological models are constructed to model how a contagion is spreading through a population, and one of the earliest models was made by Sir Ronald Ross in 1916 [41]. Often the contagion modeled is an actual contagion, i.e. some infectious, physical disease, and recently, such models have been used to model the COVID-19 pandemic, e.g. [3]. Now, to model the spread of misinformation in social networks using an epidemiological model, the contagion is exactly misinformation.

With misinformation as the contagion, one can create new representations for the classes/compartments. The susceptible class consists of people who are susceptible to believe in misinformation. The infected class consists of people who believe in misinformation and try to spread it to other susceptible people. The recovered class consists of people who formerly believed in the misinformation. The exposed class describes people who have been in contact with the contagion but have not decided whether they believe in it or not. People belonging to the known class, deliberately spread information countering the contagion (the opposite of infected). One major difficulty is how to build the model and what type of classes to include, such that they realistically represent how misinformation "infects" people and moves through the network.

2.4.1 Misinformation and the SIR/SKIR/SEIR-models

The SIR/SEIR/SKIR-models (S: susceptible, I: infected, R: recovered, E: exposed, K: known) are among the simplest compartmental models. Having models with more compartments gives more parameters which one can fit to be able to realistically model the spread of misinformation accurately. However, increasing the number of compartments will also put more constraints on the model, making it hard to adapt to different situations [36] (this will be discussed in Section 2.5). Which is why I focused on literature using simple, but general compartmental models. The selection of papers below uses simple compartmental models as a foundation; on top of this, they add complexities such as multi-layer networks or parameters and equations representing user psychology and behavior. They are diverse in their approaches and thus represent a wide variety of epidemiological models.

In "Misinformation spreading on correlated multiplex networks," Xian et al. use a network consisting of two layers (a duplex) together with a simple SIR epidemiological model, to model rumors spreading from one network to another and how they affect each other [49]. The two layers in the network are connected in such a way that all agents in one network exist in the other network, the difference being that the number of connections and where the connections are going are not equal in the two networks. This model is then used to calculate the outbreak threshold for misinformation spreading through the entire system, i.e. finding the threshold for which the system goes through a phase transition. They conclude that networks with a higher average degree (average number of connections in the system), more heterogeneous networks, and networks with positive inter-layer correlation will reduce

the outbreak threshold and thus make it easier for misinformation to infect the entire network. A heterogeneous network is a network where the degree of different nodes is diverse, such that two randomly picked nodes are unlikely to have the same degree. A positive inter-layer correlation is interpreted as if one node has a high degree in one layer of the network, it will likely have a high degree in the other layer as well. Thus, negative inter-layer connectivity represents that a node having a high degree in one layer will have a low degree in the other layer.

Chen Xu et al. also study the spread of fake news on multiple platforms and how one might contain a fake news contagion [51]. They use a simple SIR model backed by empirical data consisting of misinformation about COVID-19 spread on Facebook, Telegram, 4Chan and Gab. They propose digital vaccination (e.g. posting messages countering misinformation) as a tool to contain the contagion and found that it is dependent on how the multiplex of networks is connected. Specifically, the probability for re-entrant infection (where the contagion is removed from platform A, but re-emerges through a link from another platform) is high for multiplex networks with high connectivity between them. Thus, they conclude that using correcting information on platforms is not guaranteed to succeed if the platform is not taking action to limit the re-entrance of the contagion.

Another paper is "Rumor propagation dynamic model based on evolutionary game and anti-rumor" by Yunpeng Xiao et al. [50]. In this paper, rumors are used in place of misinformation. In short, the authors gather data about four topics from Chinese Weibo. They then create an SKIR-model and apply this model to the gathered data, and claim to be able to realistically model the spread of rumors in a network. The SKIR-model is a model where the agents with label K are agents who purposefully spread true information specifically countering the misinformation. With this very general model as a skeleton, they add additional rules for the agents in the network. Behavioral factors, psychological factors, and external factors. These factors represent numerical properties of the agents, such as user activity, number of followers, influence of rumors and anti-rumors, activity of posts, and information popularity. Thus, despite the names, they are not properties representing the behavior or psychology of users directly. Xiao et al. conclude that they have created a model able to realistically model the propagation of rumors and anti-rumors, and that it reflects the influence rumors have on agents' psychology.

In "Spreading of fake news, competence and learning: kinetic modeling and numerical approximation," Franceschi uses a basic SEIR-model to study the spread of fake news [16]. In addition, they characterize all agents by their literacy proficiency, based on a survey done by OECD [31]. The survey assesses people based on their information-processing skills, divides the participants into six levels, and examines its relation to the labor market and social outcomes. Based on the empirical data, they fit a log-normal-like distribution used for agents' competency, which they add as a Boltzmann interaction term with the usual ODEs in an epidemiological model. When agents interact with each other, they are able to lose or gain competence based on the intrinsic competences they had before the interaction. From this, a learning rate for competence is defined. By using literacy proficiency together with regular epidemiological model dynamics, Franceschi creates a stochastic model, such that transport theory and the Fokker-Planck equations are needed to solve the problem.

They show that the lower the literacy proficiency, the higher the chance of being infected by misinformation.

2.4.2 Summary and Discussion

The study of outbreak thresholds is a commonality between most of the epidemiological models I found when researching this topic. If one considers people who believe in misinformation as infected, then the breakout threshold is closely linked to the probability one infected individual has of infecting their neighbors, and the rate with which an infected person "recovers" from the misinformation [24], i.e. believes in the truth. If the threshold is breached, almost the entire system will be infected. Finding the threshold is then very important to be able to contain the contagion.

All the studies mentioned above claim to have models backed by empirical data, except Franceschi's model [16]. Using empirical data entails downloading datasets from social media websites which contain how the network is connected, and one can analyze how information will flow through them, e.g. following hashtags on X (formerly known as Twitter). By applying an epidemiological model on a real social network, one will be able to fit the spreading parameters such that the model is able to produce results aligning with analyses of propagation patterns. However, by designing models for such a special case, one has no guarantee that it will work for other cases. This will be discussed more in detail in Section 2.5.

Another common aspect of all the papers that were found for this project is the seeming lack of cooperation between social science and psychology fields and physics, mathematics, and computer science fields. The papers that were found which presented models were often collaborations between engineering, physics, mathematics, and computer science institutes. The authors characterize the agents in their network by user behavior, psychological factors, competency, or not at all. But they do not mention any of the factors, behaviors, or effects I found in my search through social science and psychology related studies. The only major overlap that was found is fact-checking, but then the modeling papers do not mention whether the correcting information lasts or not. Polarization is not mentioned either, which was shown to heavily affect how misinformation propagates through a social network. I could not find papers leaning heavily on research done within social science. All in all, there seems to be a disconnect between the two fields.

2.5 Review study of modeling of misinformation spread

This section will summarize the large review study of fake news propagation authored by Raponi et al. [36]. They have done an extensive analysis of the state of the art of existing models, with empirical data, within misinformation research. Raponi et al.'s article will provide valuable additional information and analysis about different models and the landscape of misinformation within technical fields. They conclude that there exists no model, which they have seen, that is able to capture all the properties of realistic misinformation spreading. However, they found compartmental

models which might accurately simulate specific scenarios. They analyze four different aspects of misinformation propagation in existing literature: user behavior when met with misinformation in the model, propagation models, transition patterns between states, and how network topology affects propagation.

Raponi et al. mention several models used in literature, while only in depth discussing the strengths and weaknesses of epidemiological models. Below are the main models that they mention:

- **Epidemiological models:** Compartmental models altered from disease modeling to fit misinformation propagation. Which will be explored in detail later in this section.
- **Forest fire model:** Based on the idiom that "news spread like wildfire" [7]. The authors of the relevant paper [22], collect data from X (then known as Twitter) to create a model which is analogous to how a forest fire spreads in a vegetated area. To do this, they map variables in wildfires to variables in misinformation propagation, e.g. density of the forest is equivalent to number of followers in one agent.
- **Energy model:** A model based on the laws of thermodynamics, using transmission probabilities between nodes related to agents' interest to the piece of news, which will decay over time [21]. At each time t , an infected agent has a probability to spread false news to its neighbors which will then become infected. The attraction of agents to the news will decay over time, and an agent might then forget it becoming "recovered".
- **Information cascade model:** Based on agents' behavior when digesting news online. Infected agents only have one shot to spread misinformation to its neighbors, if not successful, the agent will never be able to infect its neighbors. However, said neighbors can become infected at a later time by their other infected neighbors. If one user sees a post on social media from their infected neighbor, by scrolling past, they will never be infected by that neighbor, but by interacting in an agreeing manner (e.g. liking, sharing etc.), they become infected themselves.

While the very last three models are not discussed in detail, the methods of the papers are discussed as all three models have been used together with different datasets to provide results. Each dataset is collected from a social media platform online. This implies that the three last models are applied only on networks resembling a social media network.

The transitions between states in the models are not the only aspect affecting the misinformation propagation. The topology of the network will also decide the spreading dynamics [36]. The contagion will spread faster in a highly clustered network, i.e. a high degree of edges per node. The authors found no network which fits all models in the studies they reviewed, as they model on datasets collected from different social network sites, e.g. Facebook, X, Weibo, etc. Each of the social networks has its own functionalities, creating different network topologies. By analyzing how

the networks behave and how information spreads through them, the authors then create their own network and apply their models on them. Then they validate their results by comparing the simulation to the collected data.

2.5.1 Review of epidemiological models compared to datasets

Raponi et al. distinguish between two types of epidemiological models, stochastic and deterministic. Both types are compartmental models, where the agents involved in the diffusion process in the network belong to different compartments or classes. In the deterministic type, the transitions between the classes are deterministic, and in the stochastic type, the transitions are probabilistic.

The datasets collected are typically collected from social network sites. How they are analyzed depends on the social media platform in question. e.g. on X, the datasets are collected by looking at activity on different topics (hashtags), and collecting data on which users interacted with different posts. On Facebook, one can analyze groups and which users belong to them and what the groups discuss and post. Both of these datasets might have different network topologies. To fit a model to such a dataset, one might add complexity by increasing the number of states an agent might have (states are different from classes) and the number of transitions possible. This will increase accuracy for the specific case the model tries to simulate, but it won't necessarily fit for other cases. In the worst-case scenario, a model might only work on the specific scenario it is adapted to fit. One must therefore sacrifice accuracy and performance for generality. Implying there is a need for research to find accurate models which do not rely on increasing complexity.

2.5.2 The time factor in misinformation propagation and datasets

Many of the models ignore the time factor. Which is, by Raponi et al., mentioned as the most significant limitation of the models. The time factor is the fact that humans, and humans' behavior, change over time. One's opinions will change, the connections one has to other people change, resulting in the interactions one has with other people changing. Many of the models existing in literature do not take into account the evolution of agents' behavior or the networks in which they exist. They model misinformation spread using a dataset frozen in time. Even if the model is a perfect fit for the misinformation propagation it simulates, it has no guarantee to fit after a certain amount of time has passed. The model might give misleading results at a later point in time. By adding this time evolution, one increases the complexity of the model.

This time factor problem becomes very clear when studying epidemiological models which are compared to datasets. As described above, the authors collected datasets from popular social media websites, most of them with thousands of users. In this paragraph, let us use the model and dataset described by Scaman et al. in "A Greedy Approach for Dynamic Control of Diffusion Processes in Networks". The authors proposed an SIS model together with dynamic resource allocation (in real time) by a network manager to control the propagation/diffusion of infected nodes in

the network. Simulations were run on real-world networks, one of them was an X subgraph extracted from 1000 ego-networks. Which is a network with one focal node (the ego) connected to its peers [5]. The resulting network consisted of 81306 nodes and 1342303 edges. They then tested different strategies on how to best control the diffusion process in the network. The paper was published in November 2015, if one assumes that the X dataset was collected the same year. That would be 10 years ago when this project is written. It is by no measure safe to assume that such a dataset collected 10 years later will resemble the one collected by Scaman et al. Yielding no guarantee for similar results today.

2.6 Distributed Hypothesis Testing in literature

In 1981 Robert Tenney and Nils Sandell Jr. published a paper proposing decentralized hypothesis testing for a distributed sensor network. In which distributed sensors made statistically independent observations and then performed likelihood ratio tests on the observations, which was found to be the optimal decision rule [46]. This was the first paper found on the topic of hypothesis testing on a network of distributed nodes. And for the next 4 decades, most of the work done was related to distributed sensor networks [1][19][42][48].

In 2014 (republished in 2018), Anusha Lalitha proposed how one can use Distributed Hypothesis Testing (DHT) together with social learning (read Section 3.2 for more details). Nodes in the network perform Bayesian updates using likelihood functions, these sensor networks can communicate and thus share their beliefs, making the network reach a consensus on a true hypothesis [26]. This paper formulated the mathematical groundwork for a Distributed Hypothesis Testing model on general directed graphs.

In 2024, Diana Riazi published three papers from her PhD thesis using a DHT model to simulate how beliefs evolve in social networks subject to misinformation [38][39][40]. This was the first, and only, time a DHT model has been used to model how misinformation, and belief in it, propagates in social networks. The mathematical groundwork of the DHT model is presented in Section 3.2. Exactly why the DHT model was chosen is discussed in Section 4.

3 Theory

This section on theory will lay down the relevant mathematical foundation for graph theory and the DHT model. The simulation results presented in Section 6 depend on both the model algorithms and setup and the networks on which they are applied. The theory is presented in two subsections, the first one will contain relevant mathematics on graph theory with random graphs, Erdős-Rényi, Barabási-Albert, and Price networks. The second part will present the mathematical identities used in the DHT model, its updating rules, and measurable quantities.

3.1 Graph theory

A social network is a network of agents or users connected by links. These networks are also called graphs, consisting of nodes or vertices connected by edges. As we have already seen in Section 2.3, the structure of graphs grants insight into how misinformation spreads. Understanding how graphs work is vital, as they are not just a tool to model misinformation, but graphs and their properties are themselves fundamental for realistic simulations.

This whole section on graph theory will present the general groundwork of graphs, and then describe Erdős-Rényi, Barabási-Albert, and Price graphs. To be able to present the relevant identities for this project, I will start with the basic definitions in graph theory and build up from there.

3.1.1 Mathematical foundation for general graphs

Before describing the specific graphs relevant for this project, general definitions and parameters applicable to graphs in general will be presented. Then, I will go through the specifics for the three graph types mentioned above.

Graph theory, a field birthed out of the Königsberg problem solved by Euler in 1735, describes networks of nodes connected by edges. The number of nodes N is the number of components in the system (network). The nodes interact through links/edges which connect them to other nodes, the number of edges will be denoted as L . A network may be called a graph, and nodes may be called vertices when formulating the mathematics. An important distinction is between undirected and directed graphs. In an undirected graph, nodes connected by an edge can interact both ways along an edge. e.g. if node i and j are connected by an undirected edge, then i can interact with node j and vice versa. In a directed graph, node i may have a directed edge *to* node j , which means that interactions can only go from i to j . i.e. node j has no influence on node i . In Figure 1, one can see two simple examples of an undirected graph and a directed graph.

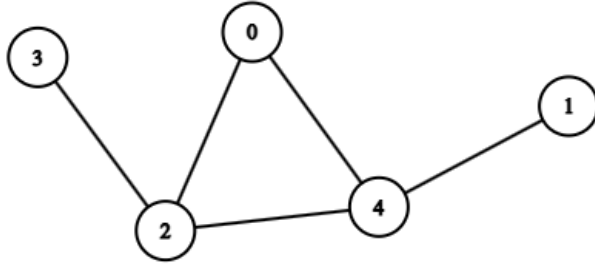
An important property for graphs is the degree, k , of nodes [6]. k_i denotes the degree of node i in an undirected graph, which is the number of edges it has to other nodes, its neighbors. For a directed graph this is denoted by k_i^{in} and k_i^{out} , representing the number of edges going out of and in to node i respectively. It is then easy to represent the total number of edges as:

$$L = \frac{1}{2} \sum_{i=1}^N k_i \quad (3.1)$$

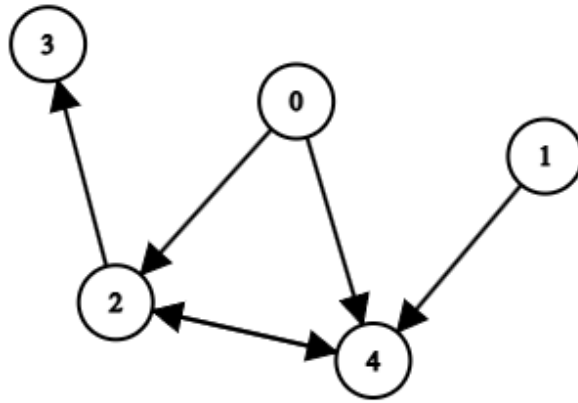
for an undirected graph, and

$$L = \sum_{i=1}^N k_i^{\text{in}} = \sum_{i=1}^N k_i^{\text{out}} \quad (3.2)$$

for a directed graph. Note how for an undirected graph, every edge is counted twice, giving rise to the factor $1/2$. This is not the case for a directed graph, as all ingoing



(a) An example of an undirected graph, with $N = 5$ and $L = 5$.



(b) An example of an undirected graph, with $N = 5$, $L = 6$.

Figure 1: A visual example of two small graphs, undirected (a) and directed (b). Note how the number of visual edges in both drawings are 5 and how still the directed graph, has more links than the undirected graph.

edges for all nodes must equal the outgoing edges. So by counting either ingoing or outgoing edges, one does not count twice and these sums must be equal. An important metric to keep track of is the average degree. Which for an undirected graph is represented as

$$\langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i = \frac{2L}{N}. \quad (3.3)$$

For a directed graph, we will distinguish between degrees in and out of a node, such that for a given node, the degree is $k_i = k_i^{\text{in}} + k_i^{\text{out}}$. The average degree will then be

$$\langle k^{\text{in}} \rangle = \frac{1}{N} \sum_{i=1}^N k_i^{\text{in}} = \langle k^{\text{out}} \rangle = \frac{1}{N} \sum_{i=1}^N k_i^{\text{out}} = \frac{L}{N}. \quad (3.4)$$

The average degree gives rise to another parameter, the degree distribution p_k . Which represents the probability that a node has degree k in an undirected graph. It can be represented as

$$\langle k \rangle = \sum_{k=0}^{\infty} k p_k. \quad (3.5)$$

For a directed graph, one must only add the superscript "in" or "out" and calculate the parameters accordingly.

A graph can be represented by an adjacency matrix A . This is a $N \times N$ matrix holding all the information about the nodes and their edges in the graph. In an unweighted graph, the matrix' elements represent:

$$A_{ij} = \begin{cases} 1, & \text{if } i \text{ is connected to } j \\ 0, & \text{if } i \text{ is not connected to } j \end{cases}. \quad (3.6)$$

Note how this adjacency matrix is defined in the same way for a directed graph as for an undirected graph. As if node i is connected to node j , only describes an edge from i to j . For an undirected edge one must add an additional requirement, namely $A_{ij} = A_{ji}$, such that edges go both ways. Which means that for an undirected graph $A = A^T$. The adjacency matrices for the graphs in Figure 1 are

$$A_{\text{undirected}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}, \quad A_{\text{directed}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}. \quad (3.7)$$

Note how A_{directed} has only 6 non-zero elements, while $A_{\text{undirected}}$ has 10. Meaning in general an adjacency matrix for directed graphs will have more redundant information than one for undirected graphs, while having more edges! Notice also how $2L_{\text{undirected}} = \sum_{i,j} A_{ij}^{\text{undirected}}$ and $L_{\text{directed}} = \sum_{i,j} A_{ij}^{\text{directed}}$. This relation will hold for unweighted graphs [6]. This will also give the following relations

$$k_i = \sum_{j=1}^N A_{ji} = \sum_{i=1}^N A_{ji} \quad (3.8)$$

for undirected graphs, and

$$k_i^{\text{in}} = \sum_{j=1}^N A_{ij}, \quad k_i^{\text{out}} = \sum_{j=1}^N A_{ji}. \quad (3.9)$$

A weighted graph is described by an adjacency matrix, such that $A_{ij} = W_{ij} \in \mathbb{R} \geq 0$.

There are several more properties of graphs, such as path length d_{ij} . Which is the shortest number of edges separating node i and j . This parameter is $d_{ij} = d_{ji}$ for undirected graphs, the equality does not hold for directed graphs. The average path length for directed networks is given by

$$d = \frac{1}{N(N-1)} \sum_{i,j=1,N;i \neq j} d_{ij}. \quad (3.10)$$

The clustering coefficient C_i for node i is a measure of the degree to which the neighbors to i are themselves connected. The clustering coefficient and the average clustering coefficient are given as

$$C_i = \frac{2L_i}{k_i(k_i - 1)}, \quad (3.11a)$$

$$\langle C \rangle = \frac{1}{N} \sum_{i=1}^N C_i. \quad (3.11b)$$

One last important result to mention is that graphs representing real networks are sparse [6]. A sparse network is a network where $L \ll L_{\text{max}} = N(N-1)/2$, i.e., the actual number of edges is much less than the maximal possible number of edges. Barabási uses a portion of the WWW with websites as nodes as an example in his book. He lists this to have $N = 325729$ and $L \approx 1.5 \cdot 10^6$, meaning this network has $L_{\text{max}} \approx 5.3 \cdot 10^{10} \gg L$.

In this project, the graphs of interest are social networks. Which are indeed real-world networks. Thus, for the rest of the paper, all networks studied will be sparse networks. In addition, the networks studied do not contain self-loops, as that is not a common artifact of social networks. A self-loop is an edge which connects a node i to itself, i.e. $A_{ii} > 0$.

3.1.2 Erdős-Rényi graphs

In this project Erdős-Rényi graphs are one of the graphs used to represent the agents and how they are connected. It is a random graph, allowing one to control the average degree of the graph which will be vital for the results produced in Section 6.2. As mentioned in Section 3.1, understanding the properties of the graphs used is crucial. To introduce Erdős-Rényi graphs, we must first define random graphs pioneered by Erdős and Rényi [14]. Let $\mathcal{G}_{N,L}$ denote all labeled graphs with N nodes and $0 \leq L \leq N(N-1)/2$ edges, then every graph $G \in \mathcal{G}_{N,L}$ has a probability

$$\mathbb{P}(G) = \left(\frac{N(N-1)}{2} \right)^{-1} \quad (3.12)$$

to be chosen. This is equivalent to starting with a graph with N nodes and 0 edges, and then creating L edges such that $\binom{N(N-1)}{L}$ are all equally likely. The graph created by this algorithm is denoted by $\mathbb{G}_{N,L}$ [17].

Gilbert defined a similar model [18]. Set $0 \leq p \leq 1$, then each graph G with N nodes and L edges has a probability

$$\mathbb{P}(G) = p^L (1 - p)^{\frac{N(N-1)}{2} - L}. \quad (3.13)$$

Again, equivalently, define a graph G with N nodes and 0 edges. Then perform $\frac{N(N-1)}{2}$ Bernoulli experiments to create independent edges with a probability p . This graph is denoted $\mathbb{G}_{N,p}$ [17].

The random graph defined by Erdős and Rényi is a uniform random graph, while Gilbert defined a binomial random graph. However, if a random graph $\mathbb{G}_{N,p}$ has L links, then it is equally likely to exist in $\mathbb{P}(G)$ defined in equation (3.12). This gives the relation

$$L = \frac{N(N-1)}{2} p. \quad (3.14)$$

Note how we have not defined if $\mathbb{G}$ is directed or undirected yet, which means the relation and definitions above hold in both cases! Equation (3.14), given the equations relating the number of edges to node degrees (3.1) and (3.2) for undirected and directed graphs respectively, results in the relations

$$\sum_{i=1}^N k_i = N(N-1)p \quad \Longleftrightarrow \quad \langle k \rangle = (N-1)p \quad (3.15)$$

for undirected graphs, and

$$\sum_{i=1}^N k_i^{\text{in}} = \sum_{i=1}^N k_i^{\text{out}} = \frac{N(N-1)}{2} p \quad \Longleftrightarrow \quad \langle k^{\text{in}} \rangle = \langle k^{\text{out}} \rangle = \frac{N-1}{2} p \quad (3.16)$$

for directed graphs. Note how for large graphs, $N \gg 1$, one can approximate $\langle k \rangle \approx Np$ and $\langle k^{\text{in}} \rangle = \langle k^{\text{out}} \rangle \approx pN/2$.

The Erdős-Rényi model for random graphs, with edge creation probability p the degree k_i for node i follows a binomial distribution [2]

$$P(k_i = k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}. \quad (3.17)$$

Which represents the number of ways k edges can be drawn from node i . There are $\binom{N-1}{k}$ indistinguishable ways of drawing k edges, and given that $i \neq j$, $P(k = k_i)$ and $P(k = k_j)$ are almost independent variables. $P(k_i = k)$ being a binomial distribution will have an expectation value of $E(X_k) = NP(k_i = k) \equiv N\lambda_k$. Where X_k denotes the number of nodes with degree k . The probability that X_k takes a given value r will approach the Poisson distribution [2]

$$P(X_k = r) = e^{-\lambda_k} \frac{\lambda_k^r}{r!}. \quad (3.18)$$

This implies that the number of nodes with degree k will follow the Poisson distribution, with expectation value $\lambda_k = NP(k_i = k)$. This expectation value is only valid if the nodes are independent, i.e. nodes that are not connected. So, the degree distribution can be approximated as a binomial distribution,

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}. \quad (3.19)$$

Now if N is large, equation (3.18) will approach a Poisson distribution with $\lambda_k = NP(k)$,

$$P(k) = e^{-pN} \frac{(pN)^k}{k!} = e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!}. \quad (3.20)$$

Where in the last equality equation (3.15) was used for large N .

The reason this degree distribution is important is the distinction of scale-free and non-scale-free networks. A scale-free network is a network in which the degree distribution follows a power law [6]. Mathematically, this means

$$P(k) \sim k^{-\gamma}. \quad (3.21)$$

Now, it is obvious that given that the degree for a large Erdős-Rényi network follows a Poisson distribution, it will not follow a power law of the form of equation (3.21). Resulting in the fact that Erdős-Rényi networks are not scale-free!

3.1.3 Barabási-Albert graphs

Many real-world social networks exhibit structures not characteristic for Erdős-Rényi graphs, but contain structures with a high degree of clustering and scale-free properties [6]. The Barabási-Albert network captures this through preferential attachment [2].

As mentioned, Barabási-Albert model is a scale-free network. To create one, one starts with m_0 nodes, with an arbitrary number of edges between them. as long as each node has at least 1 edge. Then, for every time step, add a new node to the network with $m \leq m_0$ edges. Repeat this process until the network consists of the wanted number of nodes, N [6]. To keep the model scale-free, it is important to incorporate preferential attachment, that is, the probability, $\Pi(k_i)$ that a new node connects to node i is given by

$$\Pi(k_i) = \frac{k_i}{\sum_{j=1}^{N-1} k_j}. \quad (3.22)$$

This feature ensures that new nodes have a higher probability to connect to existing nodes with a high degree compared to nodes with a lower degree. This growth of the network creates a "the rich get richer" dynamic and creates hubs (nodes with many edges) within the network.

The network grows at every time step, meaning that $N = m_0 + t$ and $L = m_0 + mt$. The sum in the denominator in equation (3.22) can then be represented as $\sum_j k_j = 2mt - m$, since it goes over all nodes except the new one. The rate at which node i grows its number of edges is

$$\frac{dk_i}{dt} = m\Pi(k_i) = \frac{k_i}{2t - 1}. \quad (3.23)$$

For large networks, i.e. large t , we get $2t - 1 \rightarrow 2t$. Which means $k_i(t)$ will grow like

$$k_i(t) = k_i(t_i) \left(\frac{t}{t_i} \right)^{\frac{1}{2}} = m \left(\frac{t}{t_i} \right)^{\frac{1}{2}}. \quad (3.24)$$

The exponent $\beta \equiv 1/2$ is the dynamical exponent. By again taking the derivative of equation (3.25) by t , one again gets the rate with which node i acquires edges

$$\frac{dk_i(t)}{dt} = \frac{m}{2\sqrt{tt_i}}. \quad (3.25)$$

To derive the exact degree distribution, we start with the number of edges expected to connect to nodes with degree k after a new node has been added. This quantity is the product of the preferential attachment, the number of nodes, the degree distribution at time t , and the number of edges created per added node. Assuming large t , we get

$$\frac{k}{2mt} NP(k, t)m = \frac{k}{2} P(k, t). \quad (3.26)$$

Where we used $N = t$ for large t . The number of nodes with degree k after adding a node $N \rightarrow N + 1$ at time $t + 1$ is the number of $k - 1$ nodes acquiring summed with the difference of the initial number of nodes with degree k and the number of nodes with degree k acquiring an edge. Mathematically:

$$(N + 1)P(k, t + 1) = \frac{k - 1}{2} P(k - 1, t) + (NP(k, t) - \frac{k}{2} P(k, t)). \quad (3.27)$$

Since all nodes added to the network start with degree $k = m$, no nodes will have degree $k < m$. Using the same arguments above, we get

$$(N + 1)P(m, t + 1) = 1 + (NP(m, t) - \frac{m}{2} P(m, t)). \quad (3.28)$$

Letting $N = t \rightarrow \infty$ and $P(k, t \rightarrow \infty) \equiv P(k)$, gives

$$P(k) = \frac{k - 1}{k + 2} P(k - 1), \quad (3.29a)$$

$$P(m) = \frac{2}{m + 2}. \quad (3.29b)$$

Using recursion and the fact that $P(k + 1) = (k/(k + 3))P(k)$, we can find the final degree distribution:

$$\begin{aligned} P(m + 1) &= \frac{m}{m + 3} P(m) = \frac{2m}{(m + 2)(m + 3)} \\ P(m + 2) &= \frac{m + 1}{m + 4} P(m + 1) = \frac{2m(m + 1)}{(m + 2)(m + 3)(m + 4)} \\ P(m + 3) &= \frac{m + 2}{m + 5} P(m + 2) = \frac{2m(m + 1)}{(m + 3)(m + 4)(m + 5)} \end{aligned} \quad (3.30)$$

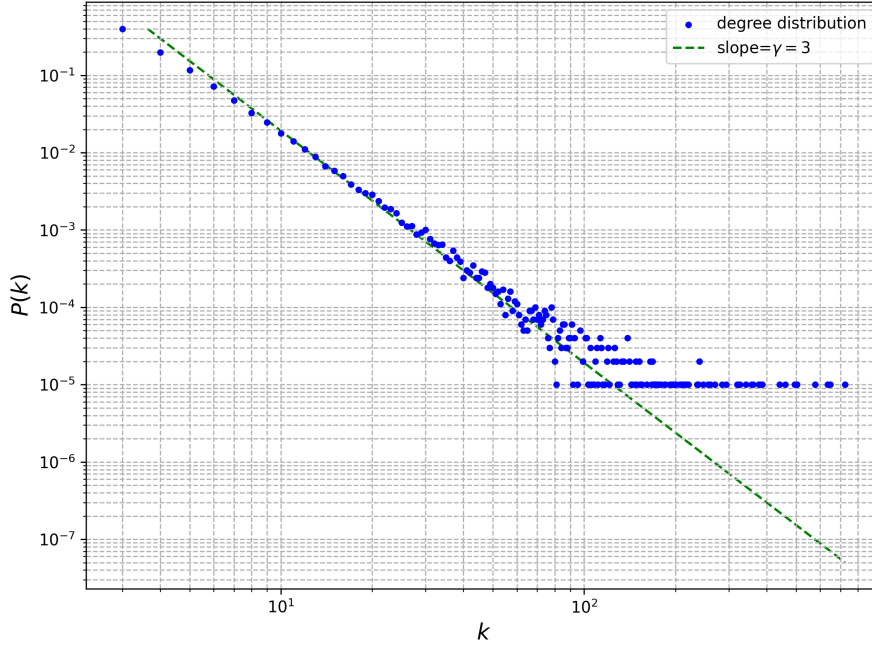


Figure 2: The power law distribution in a log-log plot for an undirected Barabási-Albert network, with $N = 10^5$, $m = 3$ and a star topology with m nodes as the initial network. The blue dots denote the degree distribution given k , and the green line is a visual aid with slope $\gamma = 3$ in this log-log plot.

By letting $k = m + 3$, the final distribution becomes:

$$P(k) = \frac{2m(m+1)}{k(k+1)(k+2)}. \quad (3.31)$$

By letting k be large, we get the result in equation (3.21). This is the degree exponent and is denoted as $\gamma = 3$.

An example of this power law can be seen in Figure 2. The distribution has a wide tail for large k , and follows the linear curve with slope $\gamma = 3$. This figure shows the fact that Barabási-Albert networks have scale-free properties.

Barabási-Albert networks are inherently undirected [6]. A directed Barabási-Albert graph (which is needed for some social networks) is the Price network, first defined in the paper "A general theory of bibliometric and other cumulative advantage processes" by Derek De Solla Price in 1976 [35]. The network is constructed in the same way as a Barabási-Albert network with one change, the preferential attachment becomes:

$$\Pi(k_i^{\text{in}}) = \frac{k_i^{\text{in}} + a_0}{\sum_{j=1}^N (k_j^{\text{in}} + a_0)}. \quad (3.32)$$

Where $a_0 \in \mathbb{R} \geq 0$ is a constant. This network will still follow a power law, but not with $\gamma = 3$, even if $a_0 = 0$. Since a newly added node i will connect to existing nodes, it will itself have $k_i^{\text{in}} = 0$, implying that when adding node $i + 1$, it will have 0% probability of connecting to node i .

Dorogovtsev et al. [12] get the set of equations:

$$\gamma = 2 + \frac{a_0}{m}, \quad (3.33a)$$

$$\beta(\gamma - 1) = 1. \quad (3.33b)$$

Which holds for all networks with preferential attachment, even those of directed types. To keep $\gamma = 3$, one requires $m = a_0$. For large γ , we expect the degree distribution to approach a small number, meaning increasing a_0 will decrease the number of nodes with a large number of edges in the network.

3.2 Distributed Hypothesis Testing

This subsection will describe the theory (math and background) for the Distributed Hypothesis Testing model. The theorems and definitions proposed by Lalitha et al. [26], on which the model is based, will not be presented in detail here. For further details, the reader is redirected to their paper "Social Learning and Distributed Hypothesis Testing" published in *IEEE Transactions on Information Theory* in 2014 and later revised and published in the same journal in 2018. First, a short historical background for the model is presented, then the math of the model and the algorithms used to simulate it will be presented. This section is strongly connected to the methods section on the same model, Section 5.1.

3.2.1 DHT model mathematical background

A Distributed Hypothesis Testing (DHT) model is a model existing on a *directed network* with N nodes, called agents. Together with this network, there exist M hypotheses connected to *one* topics, and exactly one of these hypotheses is true. The agents form a social network and are connected through L weighted edges, through which the agents interact with each other. The agents each have private and public belief vectors, which quantify their beliefs in different hypotheses. Through incoming signals and interactions with neighbors, an agent will update their belief vectors at every time step.

As mentioned above, the DHT model is applied on a directed network G with N agents and M hypotheses: θ_k . Without loss of generality, we set the true hypothesis to θ_M , which then means that θ_1 has the least correlation with the truth. In addition, we have a multivariate $N \times M$ distribution $f_i(X; \theta_{ik})$, where $i \in \mathbb{N}_1^N$ and $k \in \mathbb{N}_1^M$. $i \in \mathbb{N}_1^N$ represents that i is in all natural numbers from and with 1 up to and including N . The agents in the network will receive signals generated from this distribution. Specifically, in this model, $f_i(X; \theta_{ik})$ is a Bernoulli distribution, and all agents draw observations from the distribution corresponding to the true hypothesis.

The directed weights of the network are denoted in such a way that $W_{ij} \in \mathbb{R}_0^1$, $i, j \in \mathbb{N}_1^N$, represents the edge going from agent j to agent i . In other words, how much influence agent i accepts from agent j . If $W_{ij} = 0$, there is no edge from j to i . W is row-stochastic, such that $\sum_{j=1}^N W_{ij} = 1$. If agent i has one incoming edge, then node j will have a large influence over the beliefs of node i .

A key assumption for this model is the fact that at least one node i is able to distinguish two hypotheses. Mathematically, the Kullback-Leibler (KL) divergence for a pair of hypotheses $k \neq j$, $D(f_i(\cdot; \theta_{ik}) || f_i(\cdot; \theta_{ij}))$ is strictly positive. A positive KL-divergence implies that the system will converge to a state in which all agents believe in θ_M exponentially fast [26]. In this model, agents will confuse what observations indicate, e.g. an agent might make an observation indicating the true hypothesis but might conclude that it indicates the false hypothesis. Either way, all agents are able to distinguish the different hypotheses from each other. KL-divergence is given by the equation [25]

$$D(f_i(\cdot; \theta_{ik}) || f_i(\cdot; \theta_{ij})) = \sum_{\theta_{ik}} f_i(\cdot; \theta_{ik}) \ln \frac{f_i(\cdot; \theta_{ik})}{f_i(\cdot; \theta_{ij})}. \quad (3.34)$$

It is a measure of the relative entropy of $f_i(\cdot; \theta_{ik})$ and $f_i(\cdot; \theta_{ij})$. Riazi, in her model, implemented a system where agents group hypotheses two and two, such that they are incapable of distinguishing between two hypotheses. While ensuring at least one agent is able to distinguish between the two.

Each agent has a private belief vector $\mathbf{q}_i = (\theta_1, \dots, \theta_M)$. This vector is normalized such that $\sum_k \mathbf{q}_i(\theta_k) = 1$. As this is a private belief vector, it will not be shared with other agents. So, each agent also has a public belief vector $\mathbf{p}_i = (\theta_1, \dots, \theta_M)$, normalized in the exact same way. This public belief vector is the vector with which agents interact with each other, and it is constructed from a Bayesian update of the private belief vector together with the observations agents make.

$$p_i^{(t)}(\theta_k) = \frac{f_i(X_i^{(t)}; \theta_{ik}) q_i^{(t-1)}(\theta_k)}{\sum_{\ell=1}^M f_i(X_i^{(t)}; \theta_{i\ell}) q_i^{(t-1)}(\theta_\ell)}. \quad (3.35)$$

Where $X_i^{(t)}$, in $f_i(X_i^{(t)}; \theta_{ik})$, is the observation agent i makes at time t . Note how p_i stays normalized. All time superscripts denote the time step of the simulation. At time-step t , the private belief vector is updated as

$$q_i^{(t)}(\theta_k) = \frac{\exp \left(\sum_{j=1}^N W_{ij} \ln p_j^{(t)}(\theta_k) \right)}{\sum_{\ell=1}^M \exp \left(\sum_{j=1}^N W_{ij} \ln p_j^{(t)}(\theta_\ell) \right)}. \quad (3.36)$$

Note how also here, the private belief vector will stay normalized.

Two measures will be important to keep track of in this model. Truthfulness $\tau \in \mathbb{R}_0^1$ and cognitive dissonance $C \in \mathbb{R}_0^1$. Truthfulness is defined as the average of all agents' belief in the true hypothesis θ_M . It is expressed as

$$\tau(t) = \frac{1}{N} \sum_{i=1}^N q_i^{(t)}(\theta_M). \quad (3.37)$$

A person might experience cognitive dissonance when holding contradicting opinions, see Section 2.2 for more information. In this model, every agent has two beliefs, a

public and a private. If these do not align, agents will experience dissonance, which is represented as

$$C_i^{(t)}(\theta_k) = |q_i^{(t)}(\theta_k) - p_i^{(t)}(\theta_k)|. \quad (3.38)$$

One can from this, also calculate an average for the whole system as with the truthfulness.

Since humans inhibit confirmation bias, explained in detail in Section 2.2, one might then expect that two agents with starkly different beliefs won't influence each other all that much. To account for this bias in interactions between agents, a selective weighting is introduced. Specifically, the weights are controlled by a sigmoid function

$$W_{ij}^{(t)} = \left(1 + \exp \left[-c(\|\mathbf{q}_i^{(t)} - \mathbf{p}_j^{(t)}\| - 0.5) \right] \right)^{-1}. \quad (3.39)$$

Where c is a constant to be able to control how important confirmation is for the agents. This is done differently from Riazi in her thesis [37]. She constructs a mega-node, i.e. a node that is outwardly connected to all agents and pushes some certain agenda (either $p(\theta_M) \approx 1$ or $p(\theta_1) \approx 1$). She then formalizes that an agent i listens to a mega-node 1 with the probability $\mathbf{q}_i \cdot \mathbf{p}_1$. However, since this project does not deal with mega-nodes, there is a need to model confirmation bias between regular agents. One could perhaps also create confirmation bias functionality in the observations agents make as well.

Conspirators are also modeled in this project. A conspirator i is an agent who always has the public belief $p_i(\theta_1 \approx 1)$ and $p_i(\theta_k \approx 0)$ for $k \in \mathbb{N}_2^M$. They are immutable agents, such that their public beliefs stay constant for all t . Also meaning that neighbors have no influence over them. Since their private belief is, well private, there is no need to define it for conspirators. Although, it might be reasonable to imagine that the private and public beliefs line up for conspirators. POLES 2021 [20][20] is a survey done to quantify the proportion of Americans believing in pseudo-science and conspiracy theories. They find that around 9% of the participants believe that the COVID-19 vaccine implants a microchip in the recipient and that around 10% believe that the earth is flat. Two almost unbelievably high numbers of people believing in two preposterous conspiracy theories. This survey will be used as a baseline when choosing the number of conspirators in the network in Section 5.

4 Why choose Distributed Hypothesis Testing?

Several models (mostly epidemiological) were described in Section 2. The creators of these models also find results that fit with empirical data, and thus can explain how misinformation propagates within their system, and possibly how to prevent it from spreading. So why not choose a version of an epidemiological model?

As was also discussed in the Section 2 was how the same models lacked generality. I.e. the constraints and complexities added to the models made them accurate for the scenario studied, but perhaps not for other scenarios [36]. This could be caused by adding more classes or compartments to the model. Since the DHT model does not have these or any equivalent parameters, it will not necessarily suffer from the

same problem. The papers I read where models were constructed modeled user behavior but, the theory used was not aligned with the relevant theory found within social science [50]. There seems to be a gap between social science and physics, mathematics, and engineering on this topic. Using the literature found within social science and applying this to an epidemiological model might still run into the problem of generality. As I would have to create constraints and complexities to fit the theories I found. Thus, using an entirely different model with an entirely different groundwork is worth studying.

When deciding on a model for the project, my supervisor and I consulted Professor Asbjørn Dyrendal at the Institute of Philosophy and Religious Science at NTNU. He has also contributed pointers to research papers within social science and has checked our hypotheses about the properties of misinformation. Together, we came to the conclusion that a DHT model might be the most interesting model for this project.

The DHT model gives the problem of misinformation a nuance which does not exist with an epidemiological model. The DHT model forces all agents in the system to have an opinion on a topic. So, instead of modeling how one piece of news will spread in a network from start to finish, the DHT model, among others, is able to model how the public opinion changes in a network where all agents have been exposed to such news. Thus, the spread of misinformation is meant as how the opinion or belief in misinformation propagates and evolves in a network.

While extensive research on misinformation has primarily utilized epidemiological models, the DHT model offers a compelling alternative. Despite its emerging application within misinformation modeling, the DHT model benefits from a theoretical foundation developed in other contexts [1][19][42][48], and a solid mathematical groundwork by Lalitha et al. for applying social learning [26]. Furthermore, DHT models are uniquely capable of simulating individual belief formation and observation-based updating. Therefore, exploring the possibilities of DHT models in this field represents a significant scientific contribution by providing a perspective on how misinformation propagates and is integrated or rejected by individual agents.

The fact that the DHT model is well established on sensor networks will also bring up several difficulties. Sensors, being devoid of emotions, biases, and individuality, are not very similar to humans. This means that adjusting how the nodes in the network receive and process information is an important aspect to be able to model misinformation propagation in a realistic way. And will probably involve an extensive amount of work and research.

DHT was chosen as the model for this project for its potential in this field. It models misinformation propagation as a matter of public opinion and not a contagion, and it has an entirely different mathematical groundwork from epidemiological models (e.g. no compartments). This might prove to be a model in which one does not have the same problems of trading accuracy for generality and vice versa.

5 Method

In general, Python was used to run simulations, together with the networkx package and of course numerical libraries such as numpy and matplotlib. The code can be found in appendix A. All simulations run have been stored in npz-files such that the data should be reproducible. The seed was set to the time in integer value, since, when averaging over many simulations, no severe outliers were found. Thus, one will be able to reproduce the results shown in Section 6. Which has been confirmed by running simulations multiple times without any results being clear outliers.

All networks produced and used have been directed networks, such that from here on $k \rightarrow k^{\text{in}}$ for notational simplicity. For all simulations $N = 100$ and $M = 4$, the reasons why are explained in 5.1.

5.1 Distributed Hypothesis Testing model setup

For the mathematics and theory behind the model, see Section 3.2. For all simulations, directed networks were utilized. The networkx package has this functionality for an Erdős-Rényi network. However, since a Barabási-Albert network is undirected, and networkx does not have a function for the Price model, the Price networks were created using the same algorithm presented by Barabási in "Network Science" [6]. This with the modification that all edges created from a newly added node i go from i to the chosen nodes, and the factor a_0 in equation (3.32). The algorithm by Barabási is presented in Section 3.1. For all Price networks $a_0 = m$ was set, so that $\gamma = 3$ and $\beta = 1/2$ according to equations (3.33), which equal the values for a Barabási-Albert network. Together with a directed network, initial private beliefs were initialized using a uniform distribution.

The likelihood functions $f_i(X_i^{(t)}; \theta_{ik})$ were created from observations. These observations were modeled in a similar way as Lalitha did in "Social Learning and Distributed Hypothesis Testing" [26]. A probability vector was created for the hypotheses,

$$v_k = \frac{1 + k}{1 + M}. \quad (5.1)$$

For $M = 4$, used in all simulations, this becomes $\mathbf{v} = (0.2, 0.4, 0.6, 0.8)$. Then, the signal X was drawn from the probability corresponding to the true hypothesis, which was 0.8. For each agent, the probability that this signal confirmed the true hypothesis was then calculated using the binomial distribution: 80% of the time, the agent believed this signal confirmed the true hypothesis ($x = 1$), and 20% of the time, they believed the opposite ($x = 0$). Then the likelihood function was created for all agents and hypotheses with the Bernoulli distribution [28]

$$f_i(X_i^{(t)}; \theta_{ik}) = v_k^{x_i^{(t)}} (1 - v_k)^{1-x_i^{(t)}}. \quad (5.2)$$

If the agent i believed the signal to confirm the true hypothesis, the likelihood function became $(0.2, 0.4, 0.6, 0.8)$, if not, it became $(0.8, 0.6, 0.4, 0.2)$.

Here a Bernoulli distribution was used, which is a key methodological difference from Riazi’s work (specifically detailed in [39], distinct from [40]) which employed a multivariate Gaussian distribution. Lalitha’s paper on the DHT model mentioned using a Bernoulli distribution [26]. The Bernoulli distribution was thought to make sense, as it creates two different likelihood functions, one with larger weight towards the false hypothesis, and one with larger weight towards the true hypothesis. These likelihood functions are noiseless, meaning that two agents with likelihood functions with larger weights towards either hypothesis have the exact same likelihood functions. This will be discussed more in Section 6.

As described in Section 3.2, there was no implementation of agents not being able to distinguish between hypotheses in this model. Rather, agents misinterpreted evidence, treating observations from the true hypothesis as confirmation of the false one. This is different from Riazi in her papers and might lead to different results.

For every time step both public, $\mathbf{p}$, and private, $\mathbf{q}$, beliefs were updated according to the equations (3.35) and (3.36), and τ and C were calculated. All four quantities were stored in arrays, these arrays included all simulations performed.

If the simulations were run without confirmation bias, weights W_{ij} were initialized in a uniformly random way, using `numpy.random.rand()` with length equal to the number of incoming edges. Then, $W_{ij} = W_{ij} / \sum_{j=1}^N W_{ij}$ was performed such that W is row-stochastic. If the simulations were run with confirmation bias, weights W_{ij} were initialized according to equation (3.39). Then, $W_{ij} = W_{ij} / \sum_{j=1}^N W_{ij}$ was performed such that W is row-stochastic.

5.1.1 Simulation setups and reasons

The following simulations were performed:

1. A standard DHT model setup, that means no conspirators and no confirmation bias. Performed to show the time evolution of truthfulness and cognitive dissonance. The simulations were performed with $N = 100$ agents, $M = 4$ hypotheses, the likelihood function described by equation (5.2), uniformly random weights, 150 time steps, and 200 independent simulations. Two types of networks were utilized:
 - (a) Sparse Erdős-Rényi networks with degree $k = 10$, initialized using the `networkx` package.
 - (b) Price model networks (a directed Barabási-Albert network) with $m = 10$ and $a_0 = m$, initialized using the method described in the first paragraph of this section.
2. A standard DHT model setup, where only the final truthfulness and cognitive dissonance of the systems were calculated. Performed to show how the connectivity of the networks impacts the behavior of the system. The simulations were performed with $N = 100$ agents, $M = 4$ hypotheses, the likelihood function described above, uniformly random weights, 150 time steps, and 200 independent simulations. Two types of networks were utilized:

- (a) Sparse Erdős-Rényi networks with degrees $k \in [1, 20]$, initialized using the networkx package.
 - (b) Price model networks (a directed Barabási-Albert network) with $m \in [1, 20]$ and $a_0 = m$, initialized using the method described in the first paragraph of this section.
3. A DHT model setup with no conspirators, but with confirmation bias in the interaction between agents, as described in equation (3.39). Performed to show the time evolution of truthfulness and cognitive dissonance. The simulations were performed with $N = 100$ agents, $M = 4$ hypotheses, the likelihood function described above, weights according to confirmation bias described above, 150 time steps and 200 independent simulations. Simulations were only run on sparse Erdős-Rényi networks with degree $k = 10$, initialized using the networkx package.
 4. A DHT model setup with conspirators, with and without confirmation bias. Performed to show the time evolution of truthfulness and cognitive dissonance, in a network with conspirators, with and without confirmation bias. The simulations were performed with $N = 100$ agents, $M = 4$ hypotheses, the likelihood function described above, 150 time steps and 200 independent simulations. Simulations were only run on sparse Erdős-Rényi networks with degree $k = 10$, initialized using the networkx package. The following weights setups were used:
 - (a) Uniformly random weights.
 - (b) Weights according to confirmation bias described in equation (3.39) with $c = 4$.
 5. A DHT model setup with conspirators, with and without confirmation bias. Only the final truthfulness and cognitive dissonance of the systems were calculated. Performed to show how the number of conspirators in the networks impacts the behavior of the system. The number of conspirators was in the range $[1, 10] \in \mathbb{N}$, or in other words $[1, 10]$ % of the agents are conspirators. The simulations were performed with $N = 100$ agents, $M = 4$ hypotheses, the likelihood function described above, 150 time steps, and 200 independent simulations. Simulations were only run on sparse Erdős-Rényi networks with degree $k = 10$, initialized using the networkx package. The following weight setups were used:
 - (a) Uniformly random weights.
 - (b) Weights according to confirmation bias described above.

The reasoning for the different parameters chosen is as follows. $M = 4$ was chosen to give a somewhat diverse set of hypotheses, while still having a significant difference between the values in $\mathbf{v}$. It is also the choice of many examples in Lalitha et al.'s paper [26] and Riazi's thesis [37]. 150 time steps for each simulation were chosen, since it allowed every single simulation to reach equilibrium tens of time steps before the final time step. 200 independent simulations were chosen to give a large sample

size, to be able to calculate a standard deviation that was not all too large. $k = 10$ and $m = 10$ gave highly connected networks, which are common for many social networks [27]. It also gave a fast convergence in the Erdős-Rényi network, showing that the model worked mathematically. This will also be discussed more in 6.1.

The graph in Figure 3, described in equation (3.39), shows the weights for confirmation bias between agents' interactions. If agent i had completely similar beliefs to agent j and completely opposite beliefs to agent h , then agent i would give the interaction with agent j more than 4 times that of the interaction with agent h .

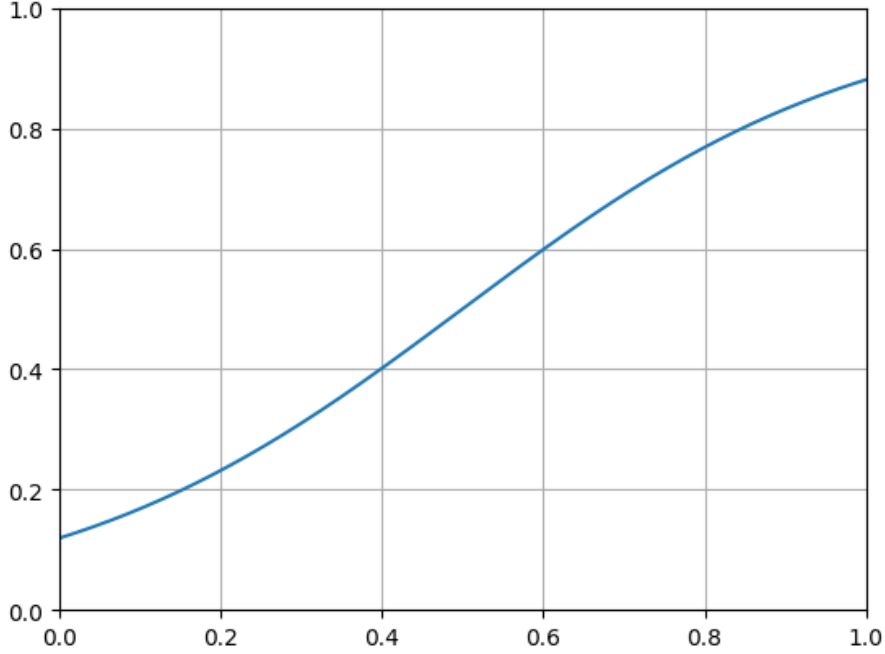


Figure 3: The sigmoid function as shown in equation (3.39) with $c = 4$. The x -axis denotes $\|\mathbf{q}_i^{(t)} - \mathbf{p}_j^{(t)}\|$ and the y -axis denotes $W_{ij}^{(t)}$.

$N = 100$ agents were chosen since it gave large enough networks for agents to have meaningful interactions. Why larger networks were not chosen are both with regards to time limitations and the purpose of the results. When timing the simulations, I noticed that the computing time does not increase linearly with N , giving much longer computation times for slightly larger N . And because of time limitations, I did not have time to simulate large networks in time for the deadline of this project thesis. Secondly, results were produced to be able to show that the model works mathematically, and that some of them are expected, with tweaks giving new results not produced before. This also includes not studying the power-law structures for Price model networks, which require networks several orders of magnitude larger than the ones used in this project.

6 Results and Discussion

An intent for this project was to build a model able to simulate the spread of misinformation and test that exact model. This section entails the implementation of the model, described in sections 5 and 3.2, and presents and discusses results produced from it. The results and discussion are split into three subsections. First, basic DHT model with no conspirators or confirmation bias. Second, how the model depends on the degree of the networks. And last, showing how conspirators and confirmation bias affect the system. The first subsection reproduces results known from Riazi’s thesis [37]. Subsection 6.4 discusses the limitations of the model and is closely linked to the future work described in the conclusion in Section 7.

6.1 Time evolution of truthfulness and cognitive dissonance

The plots in Figure 4 show the time evolution for mean truthfulness, τ , and mean cognitive dissonance, C , for a basic DHT model setup (see Section 5.1) in an Erdős-Rényi graph, i.e. no confirmation bias in neighboring interactions and no conspirators. The figures show truthfulness will quickly converge to 1 and cognitive dissonance will just as quickly converge to 0. Both are plotted with standard deviation, calculated over 200 independent simulations. This setup fulfills the assumptions for theorem 1 in Lalitha’s paper [26], which says that truthfulness should converge to 1 with probability 1 exponentially fast. This convergence is further supported by the model assumption that all agents are able to distinguish the hypotheses from each other. The plot in Figure 4a agrees with this theorem, and the basic DHT model plotted in ”Who should fight the spread of fake news?” by Riazi [40]. There is a slight discrepancy in mine and Riazi’s results. One hypothesis for the cause of this difference is that this model was implemented with a Bernoulli distribution and Riazi’s model was implemented with Gaussian distributions for the likelihood functions. Investigating how different likelihood functions for observations possibly affect results is an important point for future research (see Section 6.4).

The cognitive dissonance plot in Figure 4b converges to 0 as rapidly as the truthfulness converges to 1. This is as expected, as when truthfulness converges to 1, all agents should have private and public beliefs roughly equal and in the form $\approx (0, 0, 0, 1)$. There is also a non-zero standard deviation for cognitive dissonance after convergence.

The plots in Figure 5 show the time evolution for mean truthfulness, τ , and mean cognitive dissonance, C , for the same setup used to produce Figure 4 but using a Price network. In addition, there is a plot for the results from a system run with $k = 100$. Immediately, we see that for $k = 10$ the truthfulness does not converge to 1, it seems to find its equilibrium at around 0.4. This is in line with the results for Price networks with $k = 10$ in the appendix of [40]. Their model converges to a higher truthfulness, this might again be explained by the difference in distributions for the likelihood functions and the difference of the lack of agents unable to distinguish hypotheses in this model. For $k = 100$, i.e. a fully connected network, the truthfulness converges to 1, and cognitive dissonance to 0 in a similar manner as in Figure 4. This is also in line with the results obtained by Riazi for a Price network with $k = 100$ in

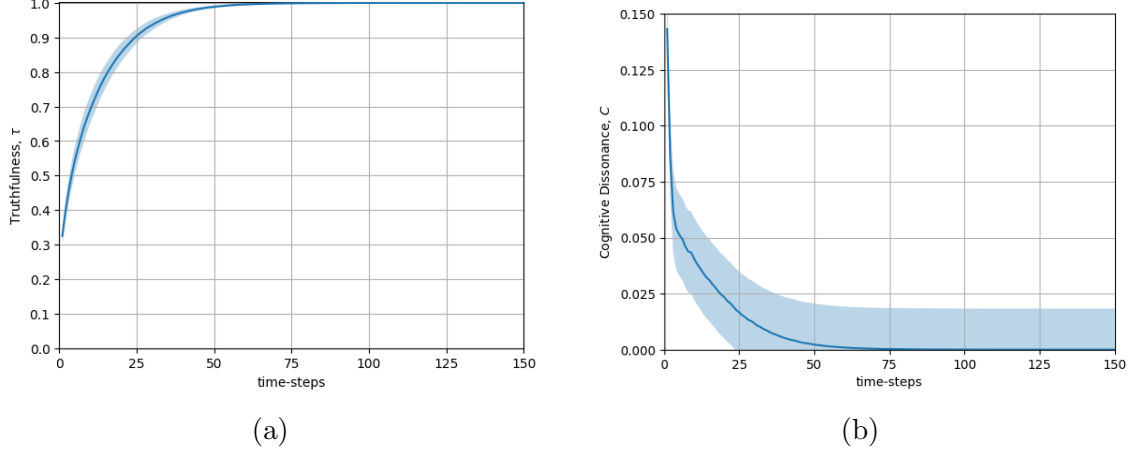


Figure 4: The two figures show the time evolution for mean truthfulness τ (a), and mean cognitive dissonance C (b) in a social network, using a basic DHT model setup on a Erdős-Rényi network. The shaded areas are the standard deviation for each parameter. The simulations were run according to the setup described in item 1a in Section 5.1.

[40].

Truthfulness likely doesn't converge to 1 for $k=10$ because the network isn't sufficiently connected. In Price networks, which exhibit a higher degree of clustering than Erdős-Rényi networks (see Section 3.1), the connectivity is insufficient to allow the widespread propagation of the true hypothesis outside local clusters. This allows pockets of incorrect beliefs to persist, thereby impeding global consensus and preventing truthfulness from reaching 1.

6.2 Truthfulness and cognitive dissonance dependence on degree

The plots in Figure 6 show the mean final truthfulness and cognitive dissonance as functions of k for an Erdős-Rényi network with a basic DHT setup (see 2a). They show that the final truthfulness and cognitive dissonance of a system are dependent on the connectivity of the networks. We can see that τ and C converge to 1 and 0 only for $k > 5$. For such values of k , the error bars show that both simulations resulted in these values, i.e. $\tau \approx 1$ and $C \approx 0$.

This shows that without confirmation bias or conspirators implemented in the model, the interaction between agents will drive the public opinion towards $\tau \approx 1$. With $k \leq 5$, there will be less interaction between agents, consequently preventing agents from spreading the true hypothesis to other agents who believe in the wrong hypotheses. This might then prevent the system from converging to a point where every agent believes the true hypothesis. This shows that the observations agents make every time step are not enough to push everyone to a state where they believe in the true hypothesis.

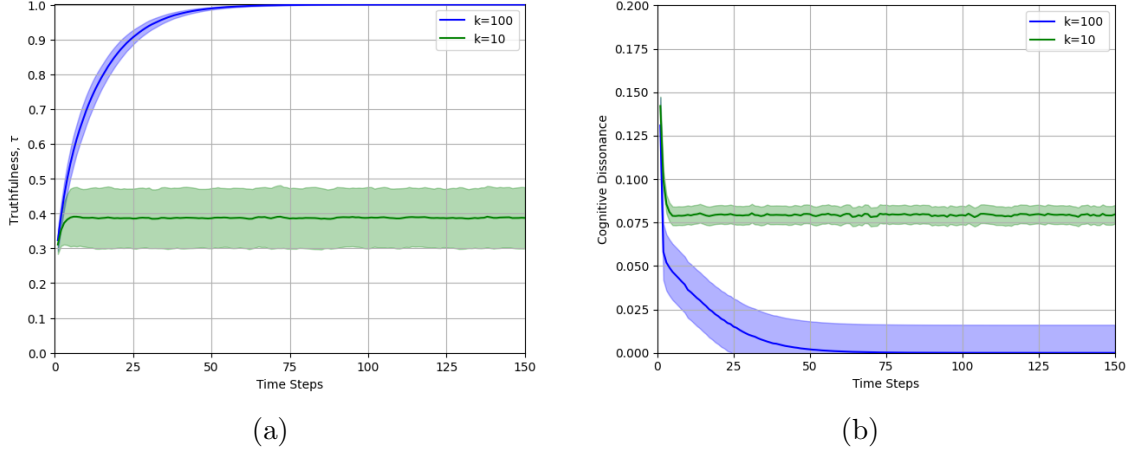


Figure 5: The two figures show the time evolution for mean truthfulness τ (a), and mean cognitive dissonance C (b) in a social network, using a basic DHT model setup on a Price network with $k = 10$ and $k = 100$. In the case of $k = 100$, the network is fully connected in this case as $N = 100$. The shaded areas are the standard deviation for each parameter. The simulations were run according to the setup described in item 1b in Section 5.1.

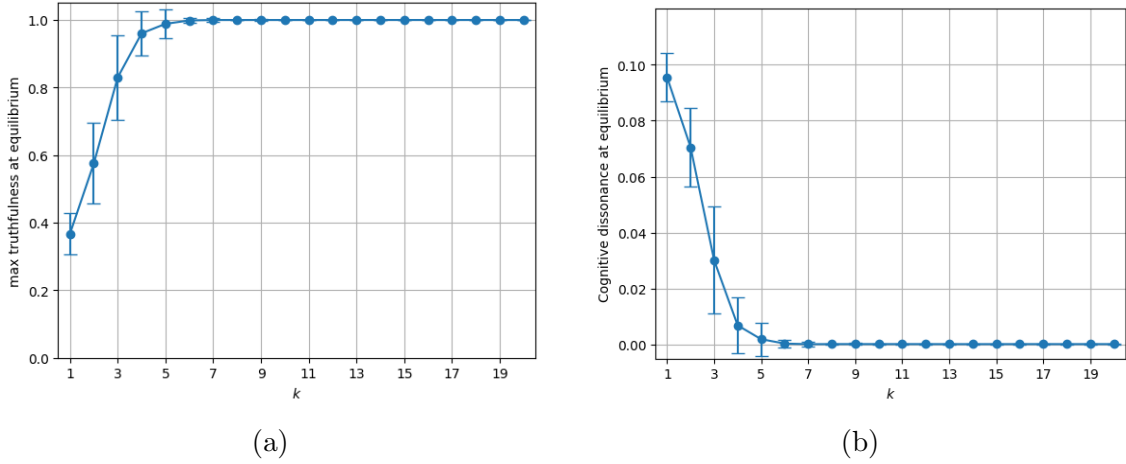


Figure 6: The mean values for truthfulness τ (a) and cognitive dissonance C (b) at equilibrium plotted against the degree $k \in [1, 20]$ of an Erdős-Rényi network. One can see that for $k > 5$, each simulation will reach $\tau \approx 1$ and $C \approx 0$. The error bars are calculated from the respective standard deviations. The simulations were run according to the setup described in item 2a in Section 5.1.

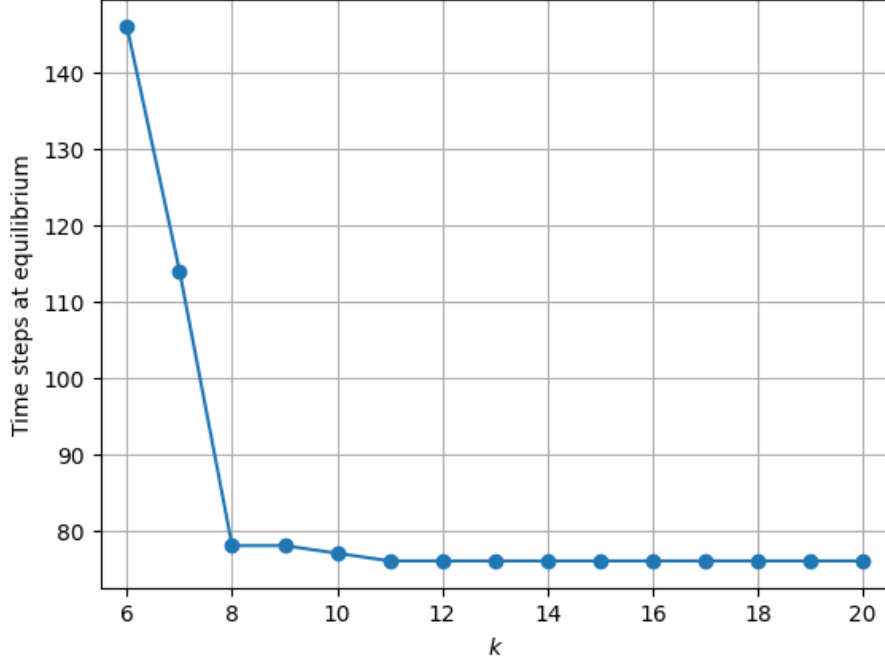


Figure 7: The number of iterations needed, on average, for each simulation to reach equilibrium. Here equilibrium is when $\tau > 0.999$. One can see that after $k = 11$, each simulation is equally fast to reach $\tau \approx 1$. The simulations were run according to the setup described in item 2a in Section 5.1.

The rate of convergence to $\tau \approx 1$ is also dependent on the degree of the network. In Figure 7 we observe that the number of time steps a simulation needs for $\tau > 0.999$ depends on k for $k < 12$. For $k \geq 11$, the number of time steps needed for truthfulness to converge to 1 remains the same. This indicates that even for highly connected networks, there is a ceiling to how fast the truthfulness will converge to 1. The plot for the Price model with $k = 100$ in Figure 5a shows that even for a network where every agent is connected to every other agent, more than 50 time steps are still needed for $\tau \approx 1$.

The picture is different for Price networks. As we saw in Figure 5a, the system with $k = 10$ was never close to reaching a state where $\tau \approx 1$. This is also the case for all $m \in \mathbb{N}_1^{20}$. In Figure 8 we see that the final truthfulness does increase with m , and since m and k are proportional (see equation 3.24), truthfulness also increases with k . However, τ increases only slightly and seems to plateau at higher m , there is also a wider spread of τ at higher m indicating more simulations with more jumbled properties. Since truthfulness does not converge to 1, cognitive dissonance does not converge to 0 either. It seems that the Price networks, which have a higher degree of clustering than Erdős-Rényi networks [2], will have a topology not allowing for τ to converge to 1 for low k . The connectivity of the agents in the network is not large enough to make the system converge to a state of complete truthfulness. The higher tendencies of clustering in Price networks might also contribute to the low truthfulness. One reason not to test for $m > 20$ is that, when creating a Price network, one must start with m nodes. Perhaps increasing N might lead to higher τ for the same m , as there will be more interactions.

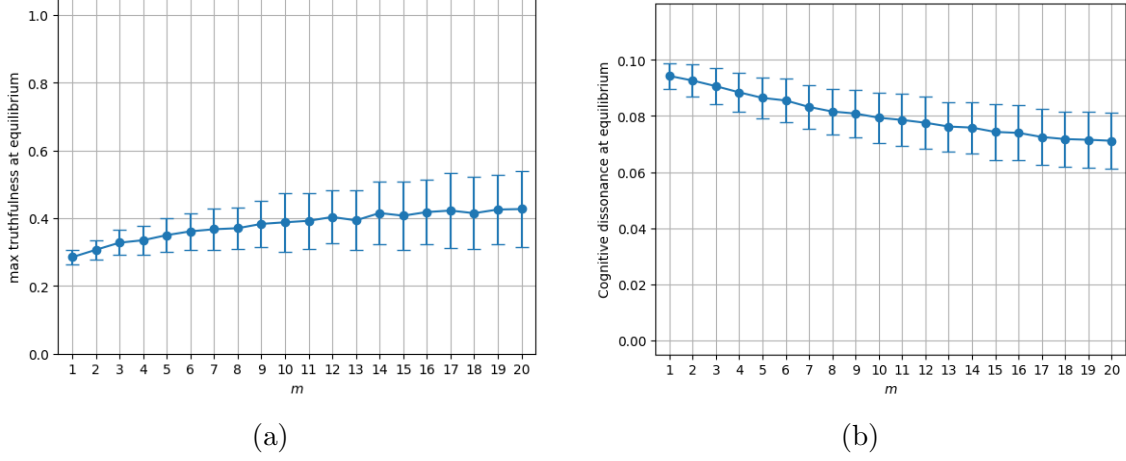


Figure 8: The mean values for truthfulness τ (a) and cognitive dissonance C (b) at equilibrium plotted against the degree $m \in [1, 20]$ of every newly added node of a Price network. The error bars are calculated from the respective standard deviations. The simulations were run according to the setup described in item 2b in Section 5.1.

Whether the truthfulness converges or not, we have seen that it is dependent on the connectivity of the networks. And that for highly connected networks, the truthfulness will converge to 1 as expected [26].

6.3 Conspirators and confirmation bias

When adding confirmation bias between agents' interactions to the model, it behaves differently. The plots in Figure 9 show how truthfulness will converge in a slower manner than a system without confirmation bias. Here, confirmation bias is modeled in a way that the system converges to a state with $\tau \approx 1$. However, by adjusting the weight of the sigmoid function used, or switching the function to model confirmation bias, one will get slower or faster convergence, or no convergence at all. Determining a confirmation bias function and strength that is based on empirical data or existing theory is an important point of study in its own right.

Figure 9b shows how cognitive dissonance will also converge slower when there is confirmation bias weighting the agents' interactions with each other.

Having conspirators in the system will heavily impact the truthfulness. The graphs in Figure 10 show the time evolution of truthfulness in a system with conspirators with and without confirmation bias in the agents' interactions. The graphs show a system in which 10 % of all agents are conspirators, see Section 3.2 and the POLES 2021 survey for more [20]. In the system without confirmation bias, truthfulness is close to 0, and in the system with confirmation bias, it is ~ 0.35 . The final value for τ in a system with confirmation bias will depend on the strength of the bias and how it is represented, either way both values for τ are extremely poor. However, the confirmation bias seems to keep some agents from believing the conspirators.

The graphs in Figure 11 show the final truthfulness as a function of the number of

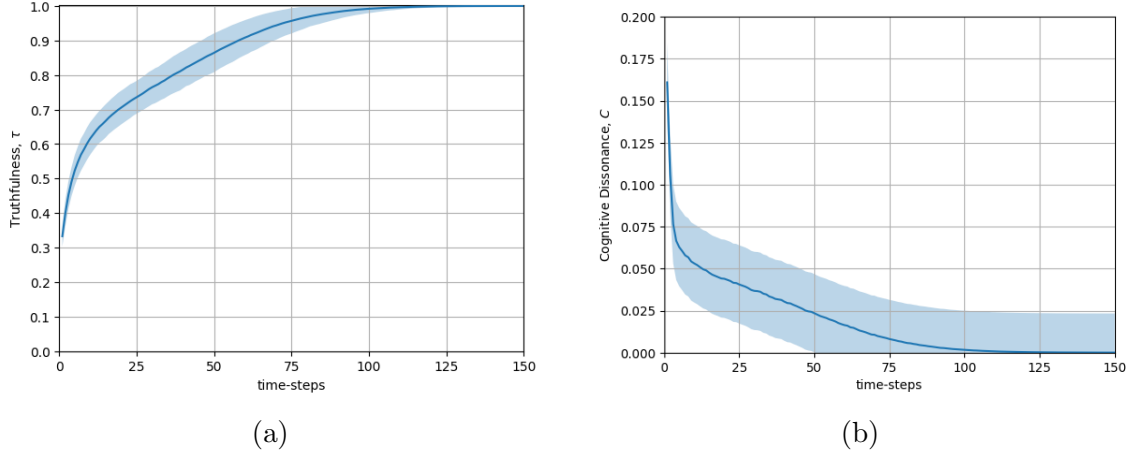


Figure 9: The two figures show the mean time evolution for truthfulness τ (a), and cognitive dissonance C (b) in a social network, using a DHT model setup with confirmation bias, as presented in Section 5.1, on a Erdős-Rényi network. The shaded areas are the standard deviation for each parameter. The simulations were run according to the setup described in item 3 in Section 5.1.

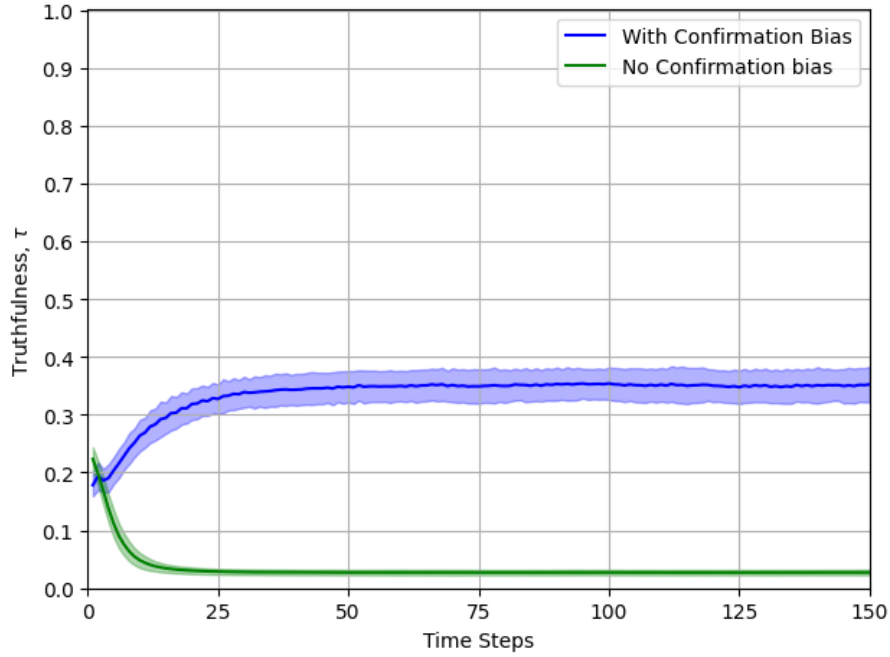


Figure 10: The time evolution of truthfulness τ , using a DHT model with and without confirmation bias, together with 10 conspirators spreading misinformation (i.e. hypothesis number 1). Simulations were run on an Erdős-Rényi network. The shaded areas are the standard deviation for each setup. The simulations were run according to the setup described in items 4a and 4b in Section 5.1.

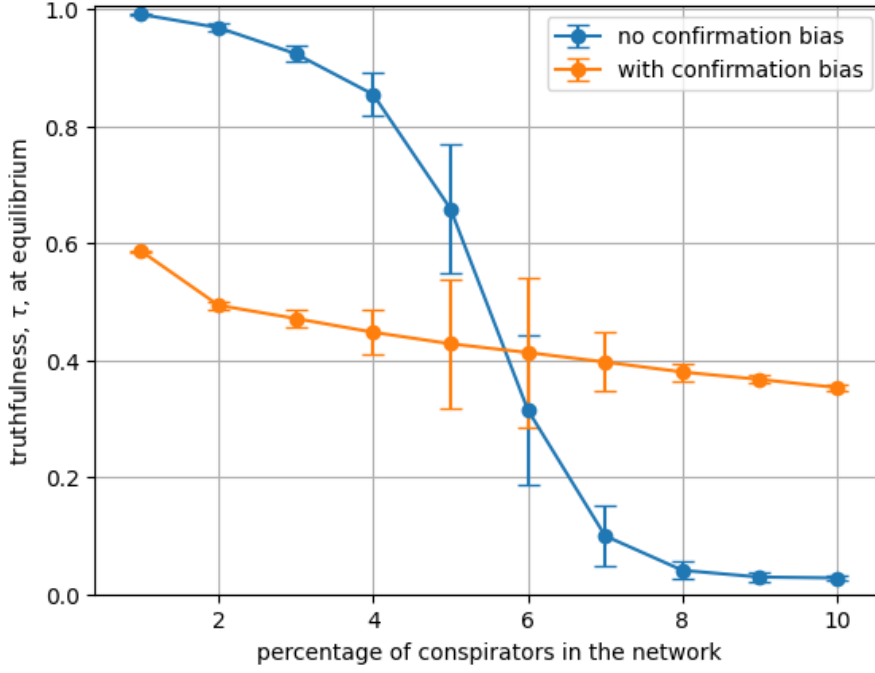


Figure 11: The truthfulness τ at equilibrium plotted against number of conspirators $\in [1, 10]$ in the network, using a DHT model with and without confirmation bias. Simulations were run on Erdős-Rényi networks. The error bars are calculated from the respective standard deviations. The simulations were run according to the setup described in items 5a and 5b in Section 5.1.

conspirators, with and without confirmation bias. Interestingly, the system without confirmation bias shows a rapid drop in truthfulness around 5% conspirators in the network. We can see that the standard deviation is larger for 5% and 6% conspirators than for other percentages, for both with and without confirmation bias. However, the system with confirmation bias has a linear decrease in truthfulness with the increase of conspirators after 2%.

Before the rapid decline of τ , the system without confirmation bias reaches the highest truthfulness, while it is the other way around after the rapid change. This might be because, early on in the simulation, a conspirator might latch onto an agent and convince them of the conspiracy before the signals or other agents convince said agent. Because of confirmation bias, this agent will be hard to turn around. Confirmation bias between agents' interactions is here shown to work with and against a system with conspirators reaching maximal truthfulness! As we can see that the lone conspirator in the system without confirmation bias almost has no effect as truthfulness is almost 1. But with confirmation bias, truthfulness is ≈ 0.6 .

6.4 Limitations

In this subsection, the limitations of the model are discussed. When implementing a model such as this, there can be multiple interesting topics to choose to study more in detail. The limitations in the implementation of this model can be points

of further research in the future. Thus, the following paragraphs will discuss the limitations of the simulations and, when possible, discuss the potential for further research.

One major limitation is the size of the networks used for simulations, especially for Price networks. With $N = 100$, the networks are too small to study the scale-free properties of Price networks. Resulting in not being able to study how misinformation spreads on a network with power-law structures. Even for Erdős-Rényi networks, it is interesting to study how the size of the network affects the results, if at all. A social network will also typically contain a number of nodes several orders of magnitude larger than $N = 100$ [6]. The networks used are not typical, realistic social networks. Which brings up a second limitation: the networks are static, while real-world social networks evolve over time. As discussed in Section 2.5, social networks are not static but evolve over time. It is a vital aspect to study how dynamic network topologies will affect the properties of misinformation spreading or beliefs changing.

Confirmation bias was introduced in the interaction between agents in the results in Section 6.3. It showed how confirmation bias makes a system with conspirators more robust, but was not based on empirical data and the strength was not varied. Using empirical data to find a fitting function and strength to represent confirmation bias is very important to research further. One will gain a deeper understanding of confirmation bias' role in and effect on the propagation of misinformation. One should also study the effect confirmation bias will have if agents exhibit it with regards to observations made. As noted in Section 5.1, the agents' observations are noiseless, meaning that if two agents interpret observations suggesting the same supporting hypothesis, their likelihood functions are identical. This might not be the best approach. Studying likelihood functions not based on the Bernoulli distribution and how they affect results is important. This is also a function that can profit from being grounded in empirical data. Lastly, modeling agents such that they also exhibit confirmation bias when making observations might be an important addition to creating realistic agents.

The final point is the non-implemented theory mentioned in Section 2.2, especially individuality and emotional arousal. How to create agents with realistic behavior patterns is difficult, but will give crucial insight into how such behaviors affect misinformation and its spread. To do this, collaborating with researchers within social science and psychology is important, as pointed out by Raponi et al. [36].

A final and significant limitation is the lack of implementation of key insights from social science and psychology, specifically individuality of and emotional arousal in agents. As discussed in Section 2.2, emotional responses and individuality are among the main drivers of belief change. Since this has not been implemented, the agents in this model behave in a relatively rational and uniform way. Developing agents with more realistic behavior patterns is difficult, but will be crucial for modeling the spread of misinformation in a realistic manner. To do this, collaborating with researchers in social science and psychology is important, as pointed out by Raponi et al. [36].

7 Conclusion

This project set out to explore the models simulating misinformation spread, and to figure out which one is the best, given research and information gathered from social science. Then using this model to simulate how misinformation propagates in social networks. While the DHT model is less frequently studied, it proved well-suited to simulate beliefs and the propagation of misinformation in social networks. The DHT model yielded expected results on highly connected Erdős-Rényi networks, where truthfulness quickly converged to 1, in alignment with the mathematics of the model and confirmed by Riazi [40]. The degree of the nodes in the network impacted the truthfulness, and for Erdős-Rényi networks $k > 5$ was needed for truthfulness to reach 1. On the other hand, Price networks, with their different structure, did not reach a state with $\tau \approx 1$ for $k \leq 20$. Implying that connectivity is not the only factor for a network to reach a state with complete truthfulness.

Introducing confirmation bias in the interactions between agents in the network slows convergence of truthfulness, showing that a well-documented psychological factor impacts how agents form beliefs. Conspirators will also have a large impact on the truthfulness. With 10% of all agents being conspirators, truthfulness was about $1/3$. Here, confirmation bias had a large positive effect, as without it truthfulness was almost 0. But even only a few percent of conspirators will have a large negative impact on the truthfulness of the system. A system with conspirators showcases an interesting rapid collapse in truthfulness around 5 – 6% conspirators. At this point, only a slight change in the number of conspirators will drastically change the truthfulness. The duality of confirmation bias will mellow this rapid collapse. For high concentrations of conspirators, confirmation bias acted as a protective mechanism, leading to comparatively higher truthfulness, whereas at low concentrations, it hindered the network from achieving high truthfulness.

The DHT model is able to simulate how beliefs evolve in social networks, and from this, it can simulate truthfulness and cognitive dissonance. It introduces nuance of belief systems, public and private beliefs, and cognitive dissonance. In contrast to epidemiological models, which use a more binary representation of believing in misinformation or not on a certain topic. The DHT model shows that it is possible to represent misinformation as something different from a contagion and shows that network structure, psychological effects/biases, and deviant agents will affect how misinformation evolves in a community. This project has its limitations. The networks used were small and static, whereas real social networks are often much larger than $N = 100$ and will evolve, even on small time scales. The agents in the network do not fully inhibit all the relevant behaviors which were emphasized in research within social science. Mainly, a realistic way to represent the individuality of agents is lacking. While confirmation bias is a well-established psychological effect, the way it is enforced in the interaction between agents is not based on empirical data. Nevertheless, its inclusion is crucial for a realistic representation of social dynamics, despite the challenges in its precise mathematical formulation. Studying how different amplitudes and functions for representing confirmation bias affects the system is missing and is important to understand. Lastly, lack of noise in the likelihood functions and the lack of agents not being able to distinguish between

hypotheses is perhaps too simple.

For the future, there is substantial work that remains to be done. Mainly, connecting the additional effects of the model, such as confirmation bias, with empirical data and established theory in social science will be important. Including additional aspects, such as the emotional arousal of content, will also be an important point of study. In general, simulating agents with realistic and representative human behaviors is vital. Producing results which might show known issues in social science, such as polarization or audience capture, will give a good indication of a realistic model. Specifically, including noise in likelihood functions and agents unable to distinguish different hypotheses from each other is more representative and might provide valuable research. Regarding the networks, studying larger-scale structures and implementing dynamic networks is essential for ensuring the model operates on realistic communities. Regarding the networks, studying larger-scale structures and implementing dynamic topologies is essential, especially for Price networks where a larger scale is crucial to investigating power-law dynamics. In general, simulating agents with realistic and representative human behaviors is key. A more robust model could be used to explore critical interventions, such as modeling how to counter misinformation.

This project emphasizes the DHT model’s potential as a powerful tool for studying misinformation. Its inherently different dynamics from other, more extensively studied epidemiological models offer a refreshing perspective. By imposing psychological effects and biases on its agents, enabling them to hold multiple beliefs, and providing a clear measure of truthfulness, the model offers a nuanced understanding of beliefs within social networks. Ultimately, the DHT model provides a useful framework for exploring how misinformation evolves and what conditions allow it to propagate effectively.

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Appendix

A Code used for this project

The code used for this project lies in the GitHub repository linked below. Inside of it, there is a README file which describes the setup of the repository.

<https://github.com/audun-sorheim/DHT-simulations-misinformation-propagation.git>